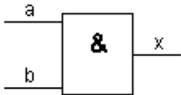
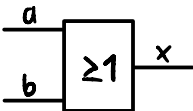
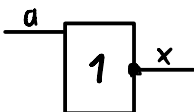
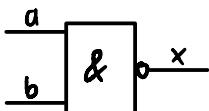
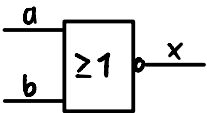
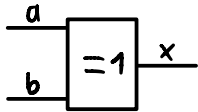
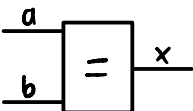


Gatter - Name, Schaltzeichen, Funktion und Wahrheitstabelle

- ① Sie haben nun im Unterricht alle wichtigen Gatter kennengelernt:
AND, OR, NOT, NAND, NOR, XOR und XNOR.

Erstellen Sie sich nun eine Übersicht, als Hilfszettel für die kommenden Stunden indem Sie die nachfolgende Tabelle ergänzen.

Name	Schaltsymbol	Funktionsgleichung	Wahrheitstabelle															
AND		$x = a \wedge b$	<table><tr><th>a</th><th>b</th><th>x</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	a	b	x	0	0	0	0	1	0	1	0	0	1	1	1
a	b	x																
0	0	0																
0	1	0																
1	0	0																
1	1	1																
OR		$x = a \vee b$	<table><tr><th>a</th><th>b</th><th>x</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	a	b	x	0	0	0	0	1	1	1	0	1	1	1	1
a	b	x																
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0	1	1																
1	0	1																
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NOT		$x = \bar{a}$	<table><tr><th>a</th><th>x</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	a	x	0	1	1	0									
a	x																	
0	1																	
1	0																	
NAND		$x = \overline{a \wedge b}$	<table><tr><th>a</th><th>b</th><th>x</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	a	b	x	0	0	1	0	1	1	1	0	1	1	1	0
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0	0	1																
0	1	1																
1	0	1																
1	1	0																
NOR		$\overline{x = a \vee b}$	<table><tr><th>a</th><th>b</th><th>x</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	a	b	x	0	0	1	0	1	0	1	0	0	1	1	0
a	b	x																
0	0	1																
0	1	0																
1	0	0																
1	1	0																
XOR		$x = a \oplus b$	<table><tr><th>a</th><th>b</th><th>x</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	a	b	x	0	0	0	0	1	1	1	0	1	1	1	0
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XNOR		$\overline{x = a \oplus b}$	<table><tr><th>a</th><th>b</th><th>x</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	a	b	x	0	0	1	0	1	0	1	0	0	1	1	1
a	b	x																
0	0	1																
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